

Final Project

Overview and Instructions

The final project constitutes **50% of the total course grade**. The goal of this project is to provide us an opportunity to develop a comprehensive understanding of data assimilation (DA) methods through theoretical study, algorithm design, implementation, and numerical experimentation. The final report is expected to demonstrate a complete DA workflow, including modeling, assimilation, evaluation, and analysis. The final report must be **typeset in L^AT_EX** and written in English in a clear, structured, and professional manner.

- **Submission Deadline and Presentation Schedule**

- First Submission (Pre-presentation): **June 8th, 11:59 PM**
- Project Presentation: June 11th (Last Lecture)
Each student will give a presentation based on their project.
- Feedback and Revision: After the presentation, feedback will be provided.
- Final Submission (Revised Report): **June 18th, 11:59 PM**
- Please upload your project report and codes via `learn.tsinghua.edu.cn`.

- **Coding Requirement:**

- Any programming language is acceptable (MATLAB and Python are preferred).
- Clearly describe your experimental setup in comments in your code.
- Generate tables and/or figures to support your findings.
- Provide discussion and interpretation of your results. For this part, please use L^AT_EX to document. Insert the tables and/or figures in the L^AT_EX generated PDF file to assist the discussion. All figures and tables must include captions, labels, and clear descriptions.

- **Use of AI Tools:**

- The use of AI tools is allowed and encouraged; briefly declare which parts used AI tools.
- You are fully responsible for all submitted content.

- **Collaboration Policy and Academic Integrity:**

- You may discuss ideas with classmates at a high level.
- All submitted work must be written independently and in your own words (can use AI tools to polish the language).
- Any external sources, including books, papers, lecture notes, online materials, and AI tools, must be properly acknowledged. All sources must be properly cited.

- **Evaluation Criteria**

Only the **final submitted version** (after revision) will be used for grading. This project is designed to help you understand how data assimilation works, not only to produce a final report. During grading, you may be asked to clarify implementation details and explain part of your report. Projects will be evaluated based on:

- Depth of understanding of data assimilation methods
- Correctness and clarity of mathematical formulation
- Quality of implementation and numerical experiments
- Proper use of DA evaluation metrics
- Completeness of the DA workflow
- Quality of analysis and discussion
- Clarity and professionalism of the report

Project Topics

Project topics are **flexible** and may include (but are not limited to):

- Application of DA methods such as:
 - Ensemble Transform Kalman Filter (ETKF)
 - Ensemble Adjustment Kalman Filter (EAKF)
 - Conditional Gaussian Nonlinear System (CGNS)
 - Yau–Yau filter
- Development of new DA methods:
 - Hybrid methods
 - Multiscale data assimilation
 - Novel extensions of existing DA frameworks
- Applications in:
 - Machine learning
 - Quantum computing
 - Engineering systems
 - Climate and geophysical modeling

Project Requirements

Each project report must include the following components (you may re-organize each of these components as sections as in a journal paper):

1. **Project Description**

Clearly describe the problem setting and the motivation of the project.

2. Background Study and Project Goals

Provide relevant background on the chosen DA method and application domain. Clearly state the objectives of the project.

3. Algorithm / Method Design

Present the data assimilation method used or developed. This includes:

- Mathematical formulation
- Algorithm design and implementation. Implement the algorithm using a programming language of your choice (e.g., MATLAB, Python). Code structure and reproducibility should be clearly documented.
- Mathematical discussion/analysis on the accuracy and stability of the DA method

4. Numerical Experiments and Results

Conduct experiments to evaluate the performance of the method. Results should include appropriate visualizations and quantitative metrics. Your project must include the following DA performance skills/metrics:

- Root Mean Square Error (RMSE)
- Pattern Correlation Coefficient (PCC)
- Entropy (for uncertainty quantification)
- Relative Entropy
- Mutual Information

5. Analysis and Discussion

Provide interpretation of results, comparison with baseline methods (if applicable), and discussion of strengths and limitations.

6. Conclusion and Outlook

Summarize your findings and suggest possible future improvements or extensions.